

Project Title : Heart Disease Prediction Using Machine Learning Models

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Course: DS_630_Machine Learning

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Abstract

This project develops a machine learning-based framework for predicting heart disease using the IEEE DataPort comprehensive heart disease dataset. The prediction task is formulated as a supervised binary classification problem where the target variable indicates the presence or absence of heart disease based on clinical and physiological attributes such as age, sex, chest pain type, cholesterol, resting blood pressure, exercise angina, oldpeak, and ST slope.

Several supervised machine learning models were implemented and compared, including Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Tuned Random Forest optimized using GridSearchCV. Data preprocessing included duplicate removal, exploratory data analysis, feature scaling for scale-sensitive algorithms, and automated feature selection using SelectKBest with the ANOVA F-test method.

Model evaluation was conducted using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC metrics. After automated feature selection, Logistic Regression achieved the best overall performance with an accuracy of 86.96% and the highest F1-score. The study demonstrates that automated feature selection can improve model interpretability while maintaining strong predictive performance for heart disease prediction.

Introduction

Heart disease remains one of the leading causes of mortality worldwide and continues to pose major healthcare challenges across both developed and developing countries. Early diagnosis is critical because delayed detection can result in severe complications such as stroke, heart failure, and sudden cardiac events.

Traditional diagnosis depends heavily on physician expertise and interpretation of multiple clinical indicators, which can become complex when several risk factors interact simultaneously. Machine learning provides a practical approach for identifying hidden patterns in clinical data and supporting predictive diagnosis through data-driven modeling.

In this project, heart disease prediction is formulated as a supervised binary classification problem where patient clinical measurements are used to predict whether heart disease is present or absent. The dataset contains demographic and physiological attributes including:

- Age
- Sex
- Chest pain type
- Resting blood pressure
- Cholesterol
- Fasting blood sugar
- Resting ECG
- Maximum heart rate
- Exercise angina
- Oldpeak
- ST slope

Multiple classification algorithms were implemented and evaluated, including:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest
- Tuned Random Forest using GridSearchCV

The objective of this project is not only to compare predictive performance but also to analyze model interpretability, robustness, and complexity in order to identify the most suitable approach for heart disease prediction.

Literature Survey

This project uses the IEEE DataPort Comprehensive Heart Disease Dataset, which combines several widely used cardiovascular datasets including Cleveland, Hungarian, Switzerland, Long Beach VA, and Statlog datasets. The combined dataset contains 1,190 patient observations and multiple clinical attributes used for cardiovascular diagnosis.

Several previous studies have explored machine learning techniques for heart disease prediction.

The first reviewed study, *Heart Disease Prediction Using Machine Learning Algorithms* by Jindal et al. (2021), compared algorithms such as Logistic Regression, KNN, and Random Forest for cardiovascular diagnosis. The study demonstrated that machine learning models can effectively support medical diagnosis when appropriate preprocessing and feature engineering are applied.

A second reviewed study, *Predictive Data Mining for Medical Diagnosis* by Soni et al. (2017), discussed classification approaches including Decision Trees, KNN, Naive Bayes, and Neural Networks. The study emphasized the importance of preprocessing, feature selection, and proper evaluation metrics in medical prediction systems.

These studies guided the methodological understanding of this project; however, all experiments and analyses in this work were independently performed using the IEEE DataPort heart disease dataset.

Machine Learning Formulation

This project is formulated as a supervised binary classification problem. Let X denote the input feature vector containing patient clinical measurements:

$$X = \{x_1, x_2, x_3, \dots, x_{11}\}$$

where the predictor variables include age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG, maximum heart rate, exercise angina, oldpeak, and ST slope.

The target variable y represents heart disease status:

$$y \in \{0, 1\}$$

where:

$y = 0 \rightarrow$ No heart disease

$y = 1 \rightarrow$ Heart disease present

The objective is to learn a classification function:

$$f(X) \rightarrow y$$

that maps clinical attributes to the predicted disease category.

For a given patient i :

$$\hat{y}_i = f(x_i)$$

where $\hat{y}_i$ is the predicted output produced by the machine learning model.

Several supervised classification models were implemented:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest

- Tuned Random Forest via GridSearchCV

Model performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Automated Feature Selection Procedure

One important correction in this project was the implementation of automated feature selection.

In the earlier version of the project, feature importance analysis was discussed, but no true automated feature selection algorithm had been applied. To correct this, the updated project uses:

SelectKBest(score_func=f_classif, k=8) \text{SelectKBest(score_func=f_classif, k=8)}

The SelectKBest algorithm automatically evaluates each feature using the ANOVA F-test and selects the top $k=8$ statistically relevant features based on their relationship with the target variable. This selection process is fully algorithmic and does not rely on manual filtering or prior domain assumptions.

The selected features were:

- age
- sex
- chest pain type
- fasting blood sugar
- max heart rate
- exercise angina
- oldpeak
- ST slope

The highest ANOVA F-scores were observed for:

- ST slope
- Exercise angina
- Chest pain type

which indicates strong predictive relevance for heart disease classification.

Model	Hyperparameters
Logistic Regression	max_iter=1000
KNN	n_neighbors=7
Decision Tree	max_depth=5, min_samples_split=10
Random Forest	n_estimators=100, max_depth=5
Tuned Random Forest	GridSearchCV with n_estimators=[50,100,200], max_depth=[3,5,7,None], cv=5

Tabulated Metrics:

Model	Main Hyperparameters	Accuracy	Precision	Recall	F1 Score
Logistic Regression	max_iter=1000; SelectKBest k=8; standardized	0.8696	0.8611	0.9118	0.8857
KNN	n_neighbors=7; SelectKBest k=8; standardized	0.8641	0.8667	0.8922	0.8792
Decision Tree	max_depth=5; min_samples_split=10; SelectKBest k=8	0.7935	0.8137	0.8137	0.8137
Random Forest	n_estimators=100; max_depth=5; SelectKBest k=8	0.8641	0.8667	0.8922	0.8792
Tuned Random Forest	GridSearchCV best params={'max_depth': 5, 'n_estimators': 50}; SelectKBest k=8	0.8641	0.8667	0.8922	0.8792

Complexity Accuracy Tradeoff Analysis

The models demonstrated different tradeoffs between predictive performance, interpretability, and computational complexity.

- Logistic Regression achieved the highest accuracy while maintaining high interpretability and low complexity.
- KNN performed competitively but has higher prediction-time computational cost because it stores all observations.
- Decision Tree provided interpretability but lower predictive stability.
- Random Forest provided strong robustness and ensemble stability.
- Tuned Random Forest did not significantly improve test performance after feature selection.

After automated feature selection, Logistic Regression became the strongest overall model.

Conclusion

This project demonstrated that machine learning can effectively predict heart disease using structured clinical variables from the IEEE DataPort comprehensive heart disease dataset.

A major improvement in the corrected version was the implementation of automated feature selection using SelectKBest with the ANOVA F-test. This ensured that feature selection was algorithmic rather than manually guided.

Several supervised machine learning models were compared, including Logistic Regression, KNN, Decision Tree, Random Forest, and Tuned Random Forest.

The experimental results showed that Logistic Regression achieved the best overall performance after automated feature selection, with:

- Accuracy = 86.96%
- Recall = 91.18%
- F1-score = 88.57%

These results indicate that the selected clinical features preserved strong predictive information and were highly effective for linear classification.

Future work may include:

- XGBoost and boosting ensembles
- Recursive feature elimination (RFE)
- Deep learning methods
- External clinical validation datasets

References

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